



Лекция 09

Проблемы обучения и подбора архитектуры

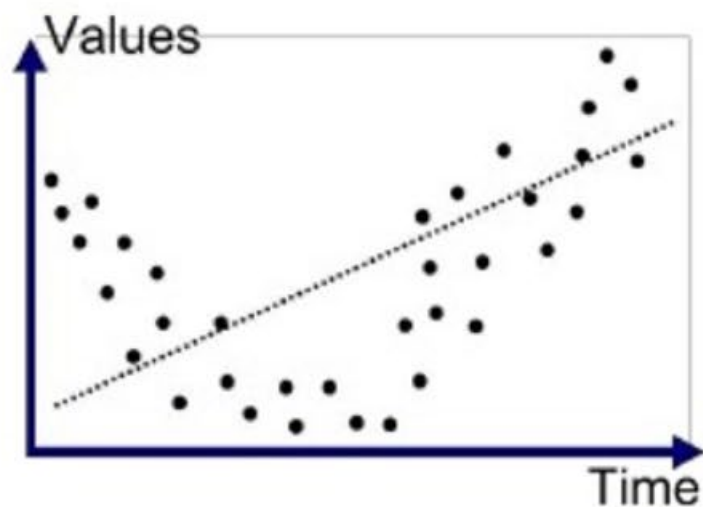
- А. Переобучение и проверка (валидация)
- Б. Регуляризация, нормализация, дропаут
- В. Подбор гиперпараметров, поиск архитектуры



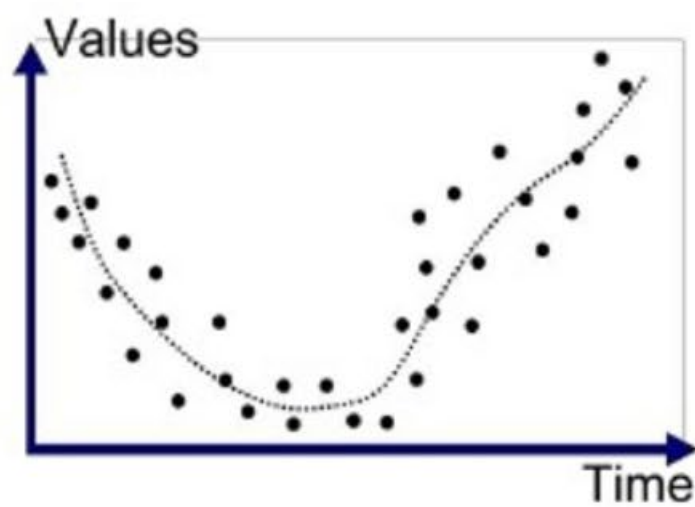
Способы повышения обобщающей способности

Недообучение и переобучение моделей

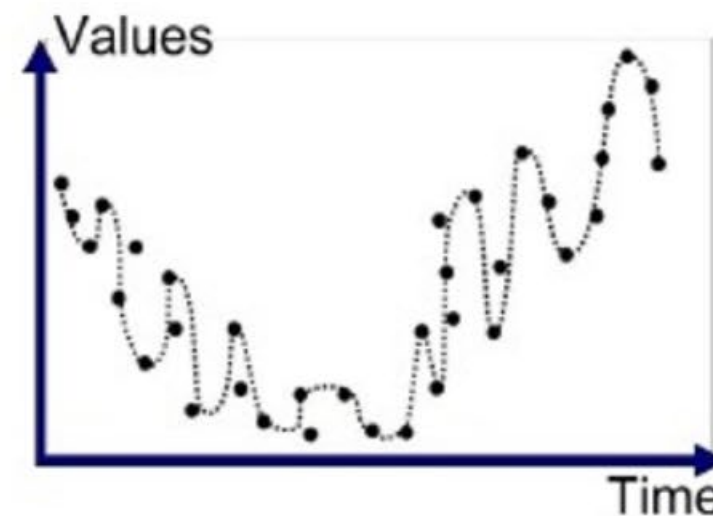
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Underfitted



Good Fit/Robust

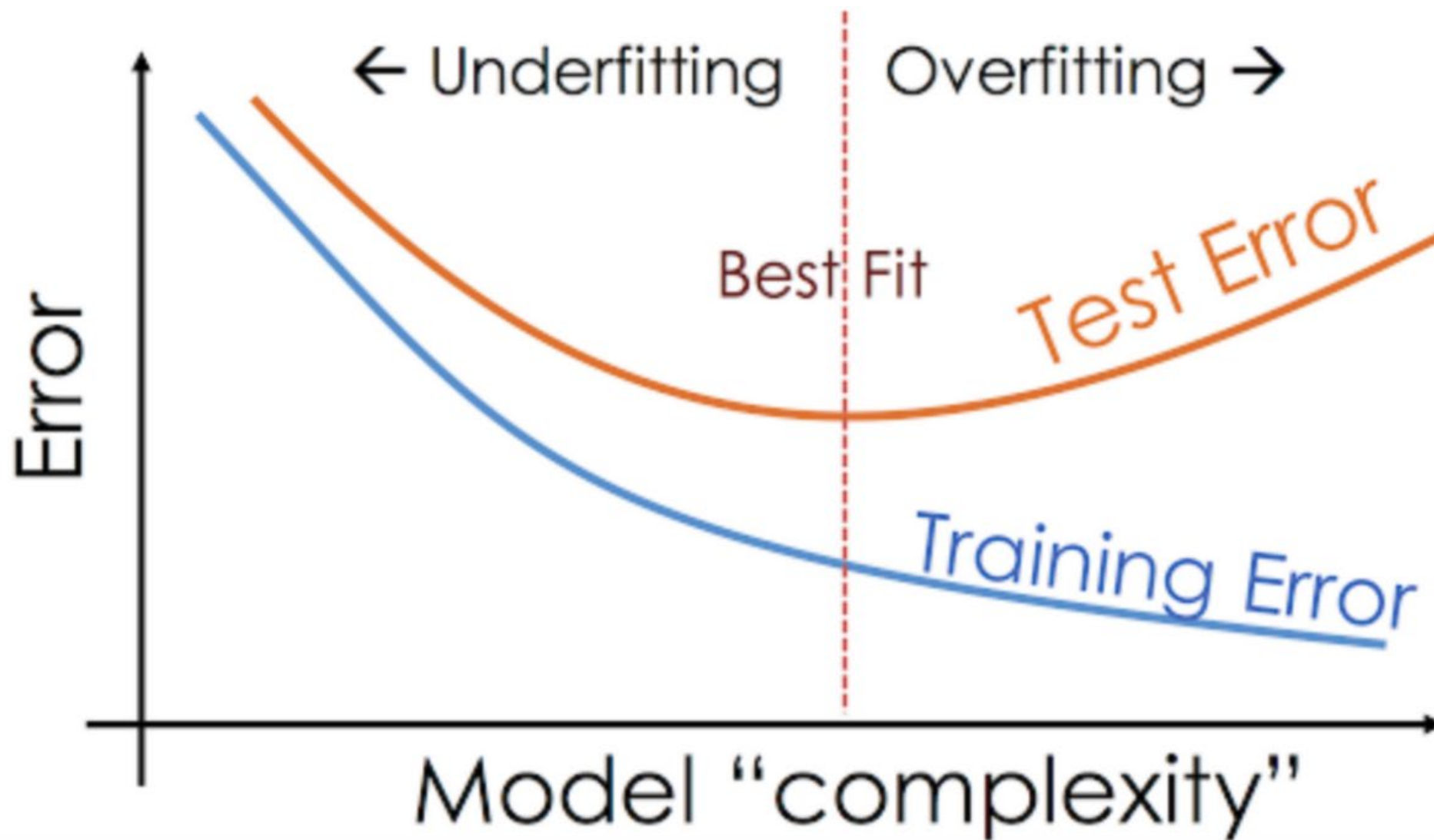


Overfitted



Сложность модели и обобщающая способность

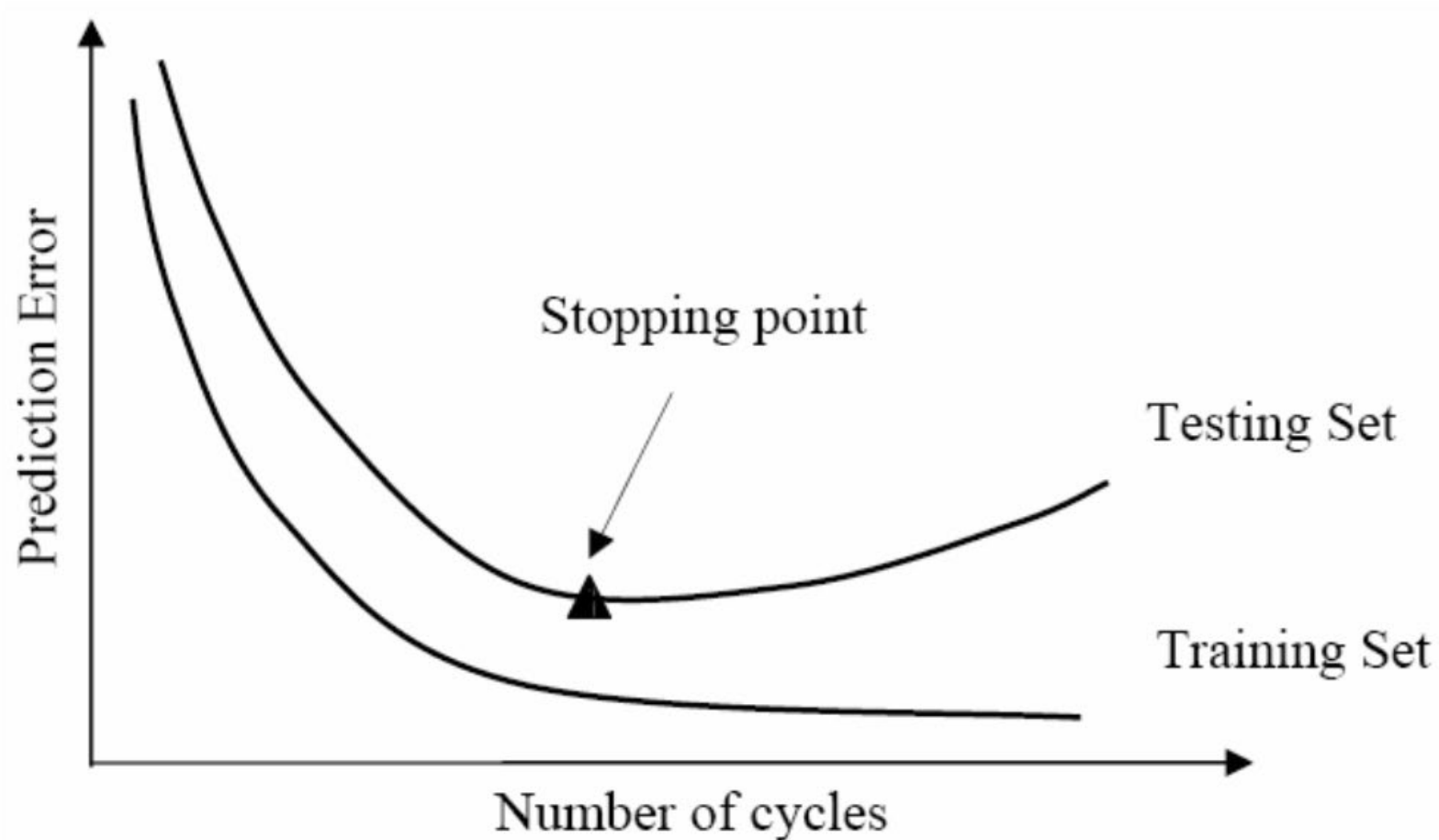
3





Различие обучающей и обобщающей способности

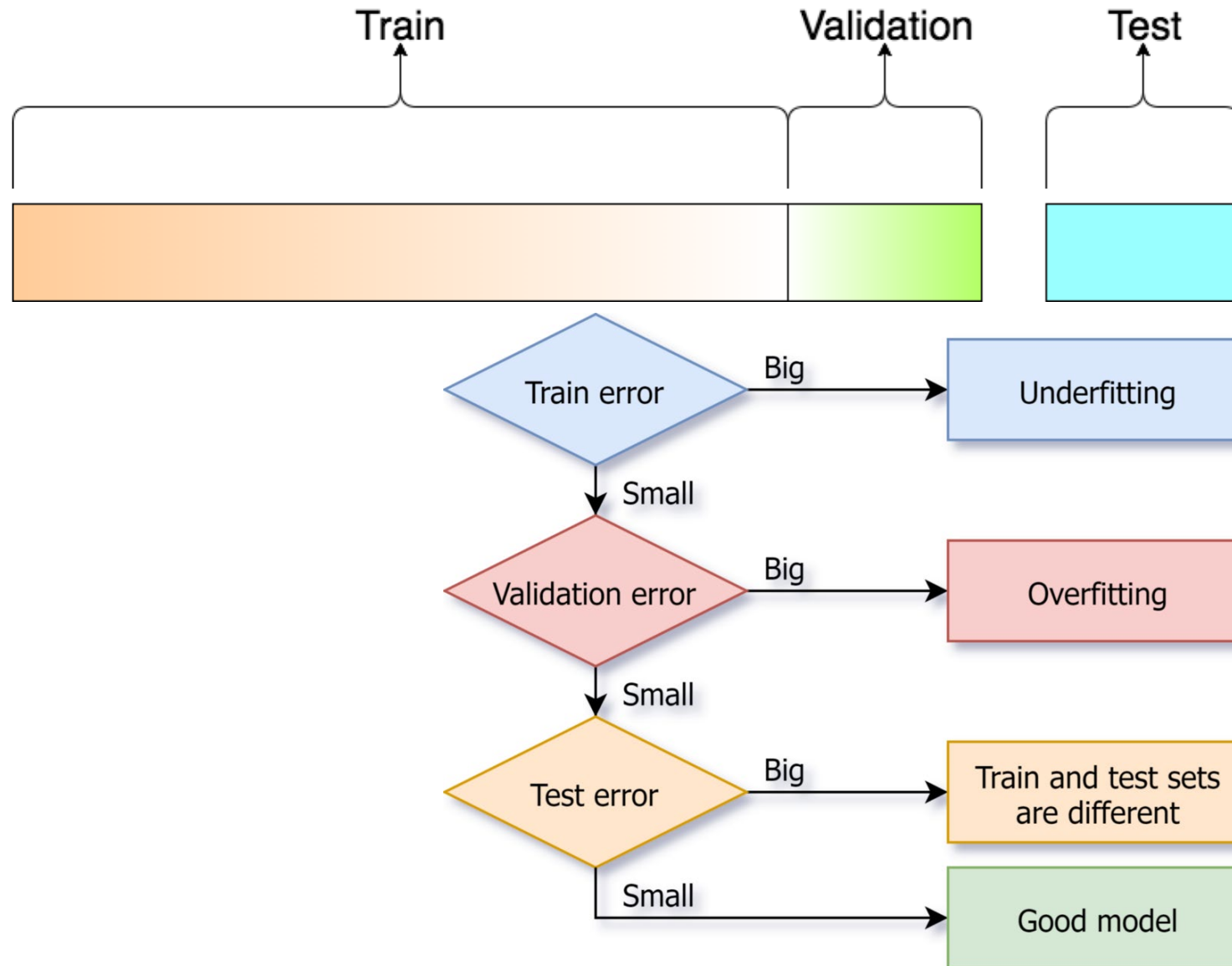
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Обучающий, тестовый и проверочный набор данных

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Борьба с недообучением и переобучением

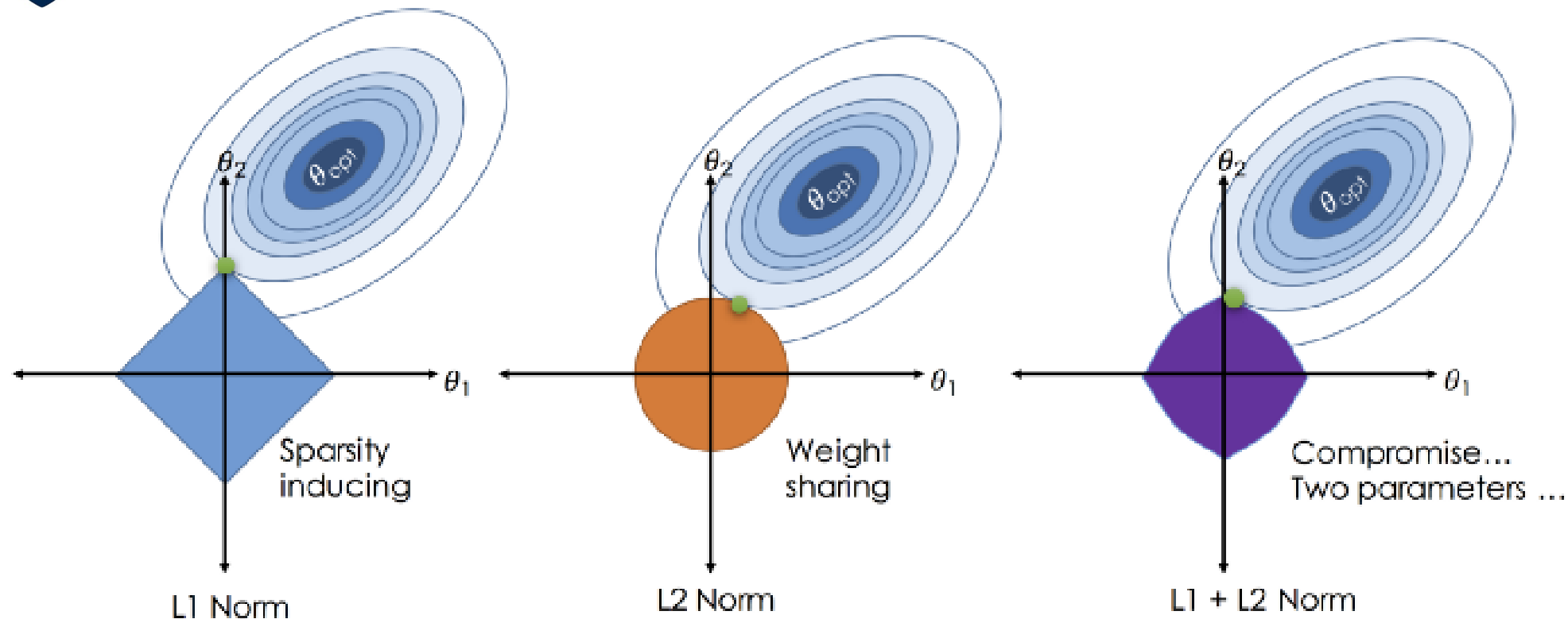
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Techniques to fight underfitting and overfitting		
	Underfitting	Overfitting
Complexity of the model	More complex model Try a more powerful model with a larger number of parameters Ensemble learning More layers / number of neurons per layer	More simple model Try a less powerful model with a fewer number of parameters Less layers / number of neurons per layer
Regularization	Less regularization Decrease regularization	More regularization Increase regularization impact Early stopping, L1 / L2 regularization, dropout
Quantity of features	A larger quantity of features Get additional features, feature engineering, polynomial features, etc.	A smaller quantity of features Remove all additional features, feature selection
Data	Data cleaning, hold-out validation or cross validation. Getting more data most likely will not help	Data cleaning, hold-out validation or cross validation. Getting more data most likely will help (data augmentation)



Регуляризация

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$$L = L_{\text{задача}} + \alpha * L_{\text{регуляризация}}$$

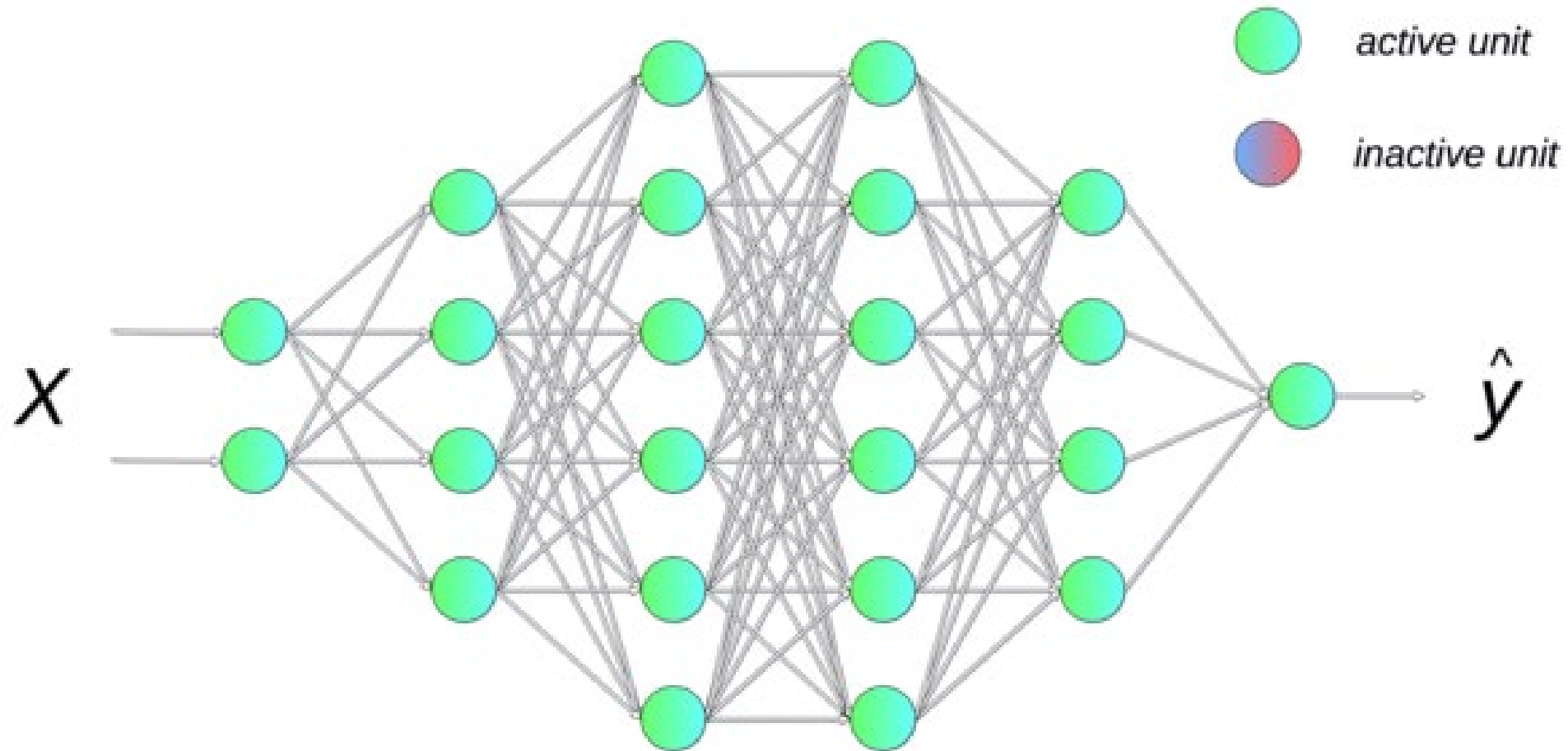
$$L_1 = \sum_i |W_i|$$

$$L_2 = \sum_i W_i^2$$



Dropout

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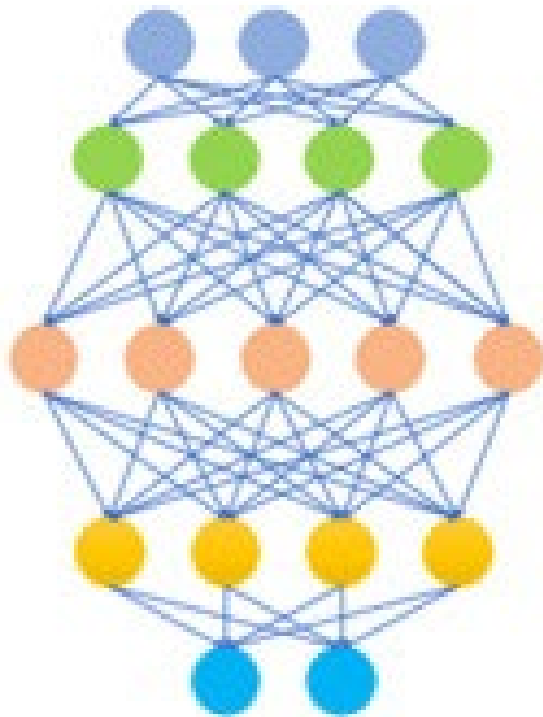
$$p^{[0]} = 0.0 \quad p^{[1]} = 0.0 \quad p^{[2]} = 0.5 \quad p^{[3]} = 0.0 \quad p^{[4]} = 0.25$$



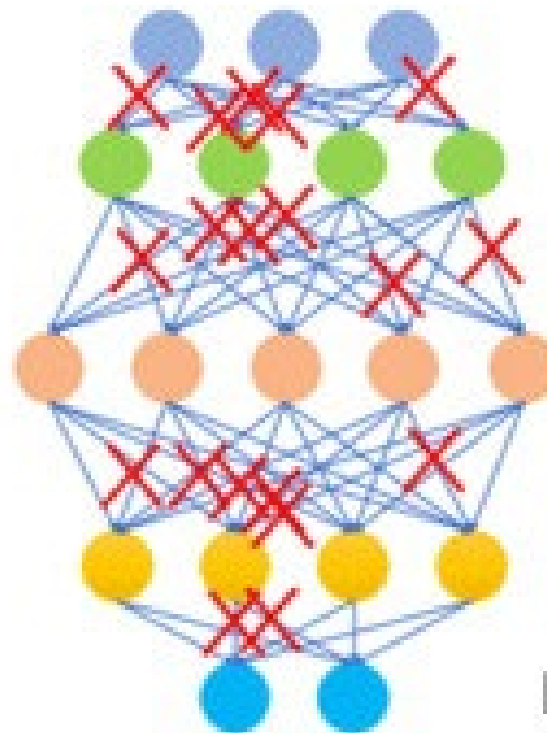
Pruning

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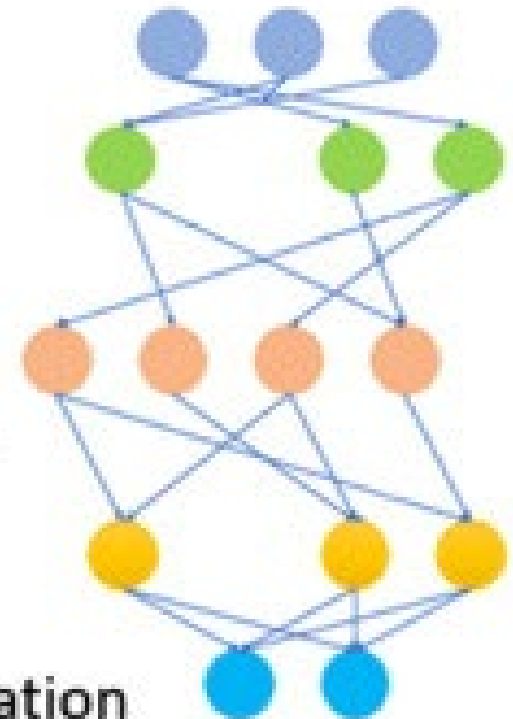
Dense network



Weight Pruning



Node Pruning



Regularization



Нормализация BatchNorm

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$$y_i = \left(\frac{x_i - \mu}{\sigma} \right) \gamma + \beta$$

Input: Values of x over a mini-batch: $\mathcal{B} = \{x_1 \dots x_m\}$;

Parameters to be learned: γ, β

Output: $\{y_i = \text{BN}_{\gamma, \beta}(x_i)\}$

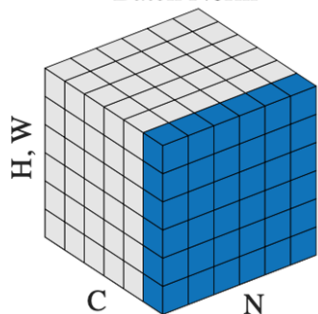
$$\mu_{\mathcal{B}} \leftarrow \frac{1}{m} \sum_{i=1}^m x_i \quad // \text{ mini-batch mean}$$

$$\sigma_{\mathcal{B}}^2 \leftarrow \frac{1}{m} \sum_{i=1}^m (x_i - \mu_{\mathcal{B}})^2 \quad // \text{ mini-batch variance}$$

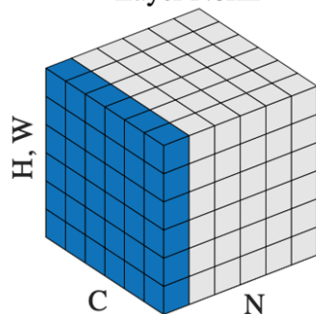
$$\hat{x}_i \leftarrow \frac{x_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \quad // \text{ normalize}$$

$$y_i \leftarrow \gamma \hat{x}_i + \beta \equiv \text{BN}_{\gamma, \beta}(x_i) \quad // \text{ scale and shift}$$

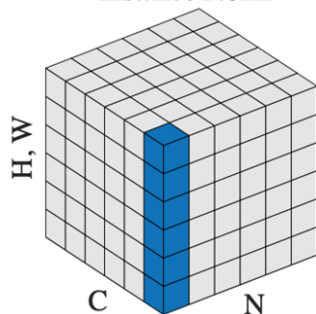
Batch Norm



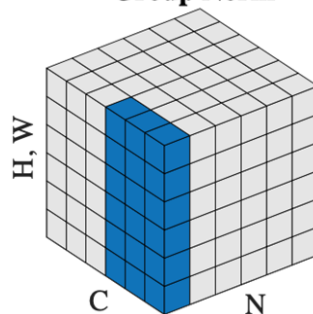
Layer Norm



Instance Norm



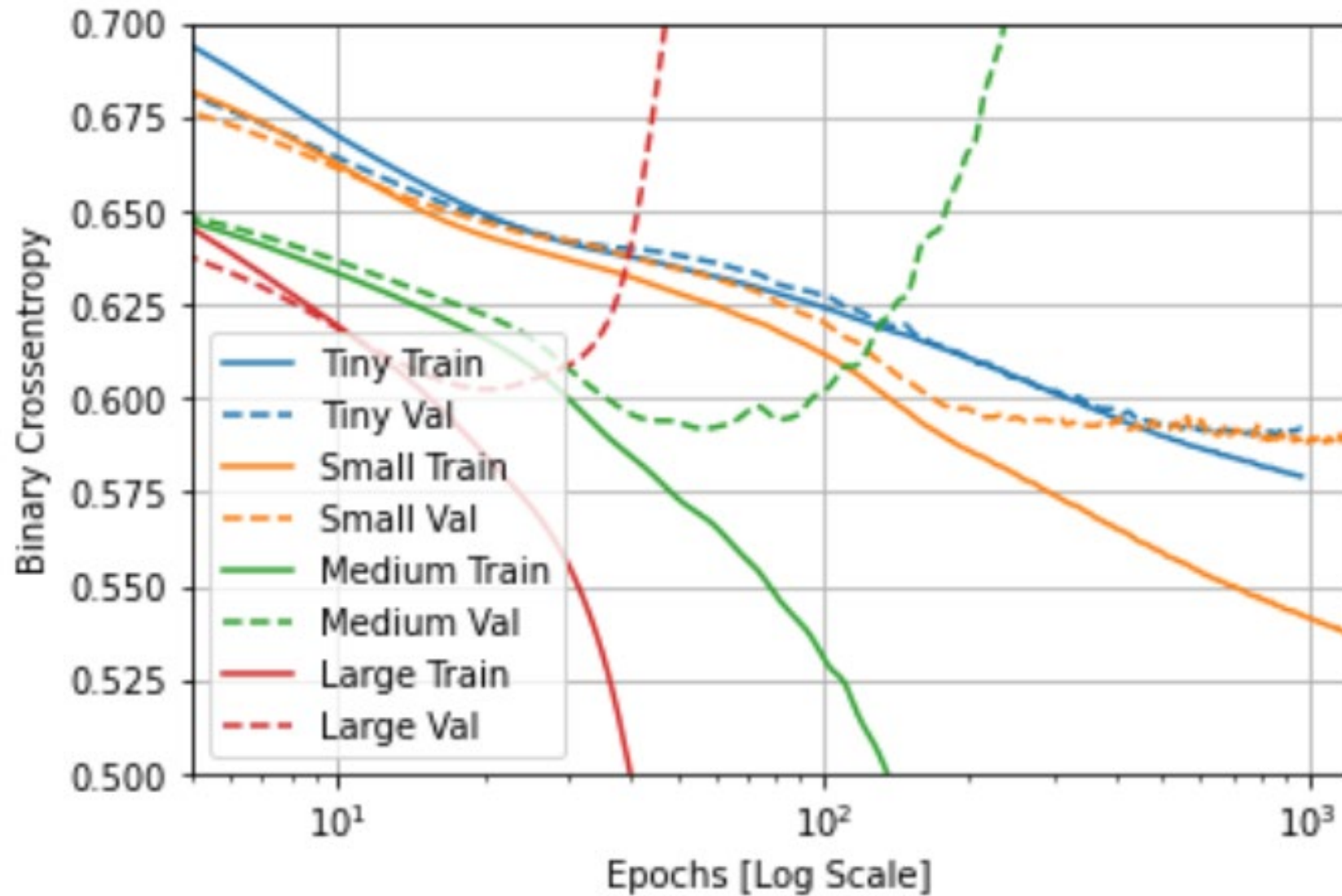
Group Norm





Примеры. Модели разного размера

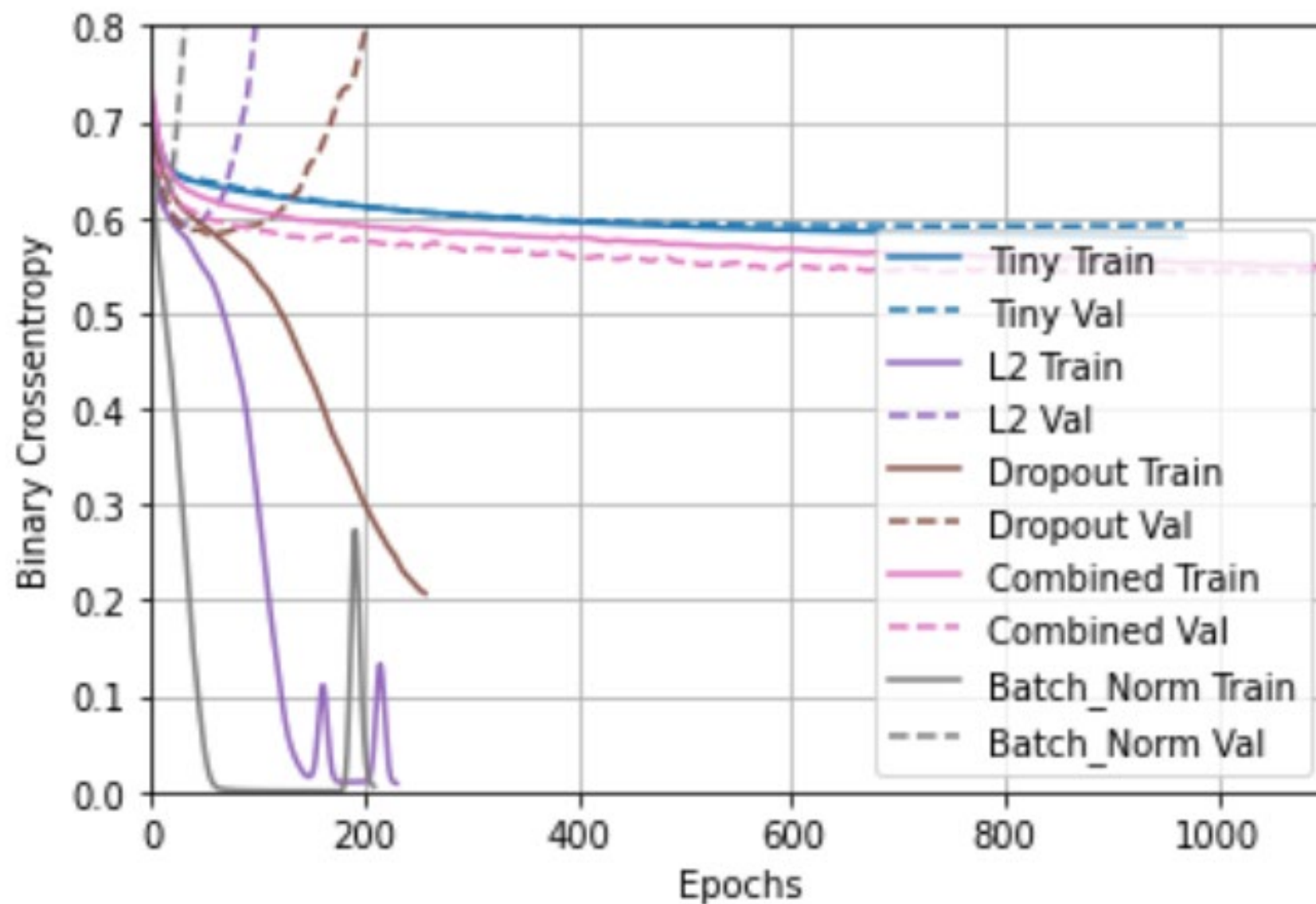
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Примеры. Нормализация

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Настройка гиперпараметров

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NON
BRAND
DATA

Non-Brand Data

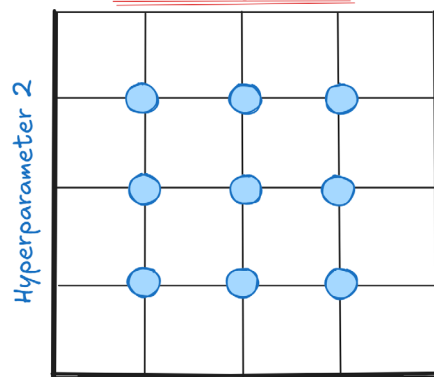


Cornellius Yudha Wijaya

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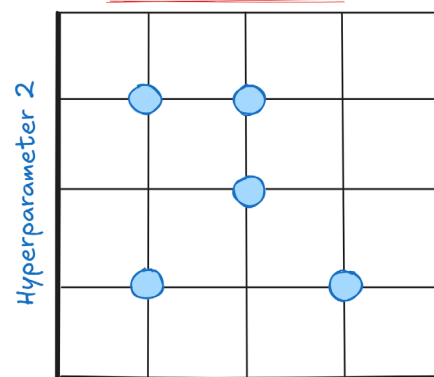
6 Common Hyperparameter Optimization Techniques

Grid Search



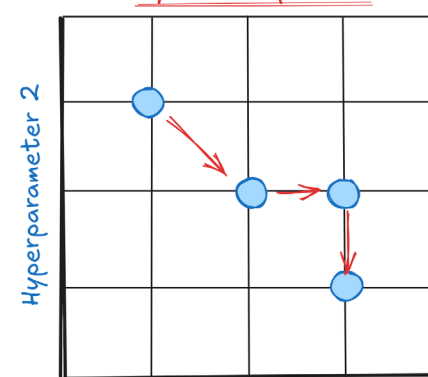
Hyperparameter 1

Random Search



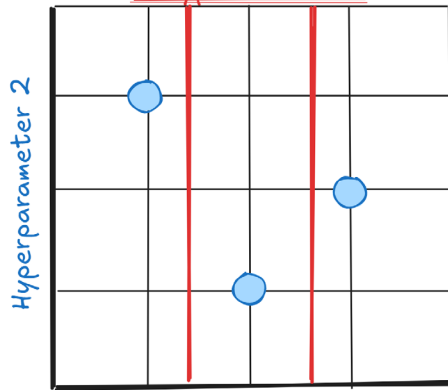
Hyperparameter 1

Bayesian Optimization



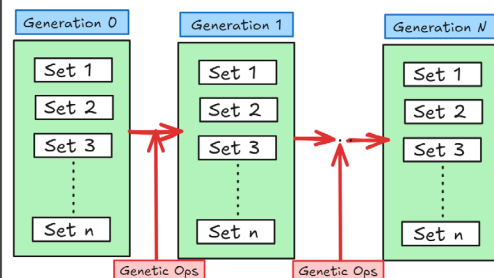
Hyperparameter 1

Hyperband

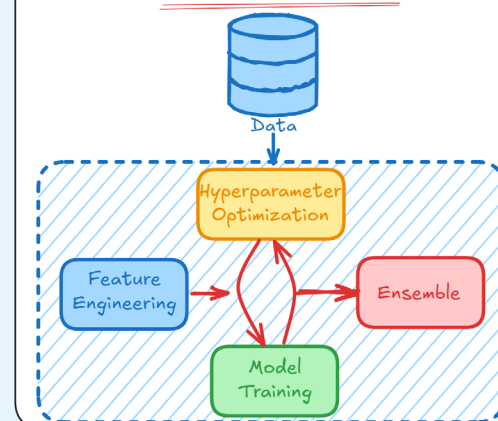


Hyperparameter 1

Genetic Algorithm



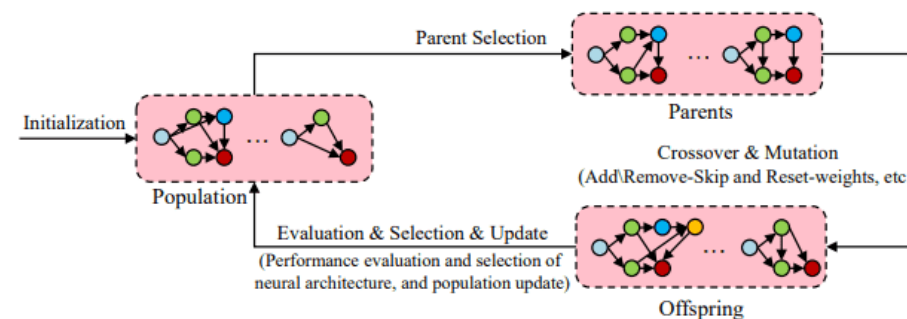
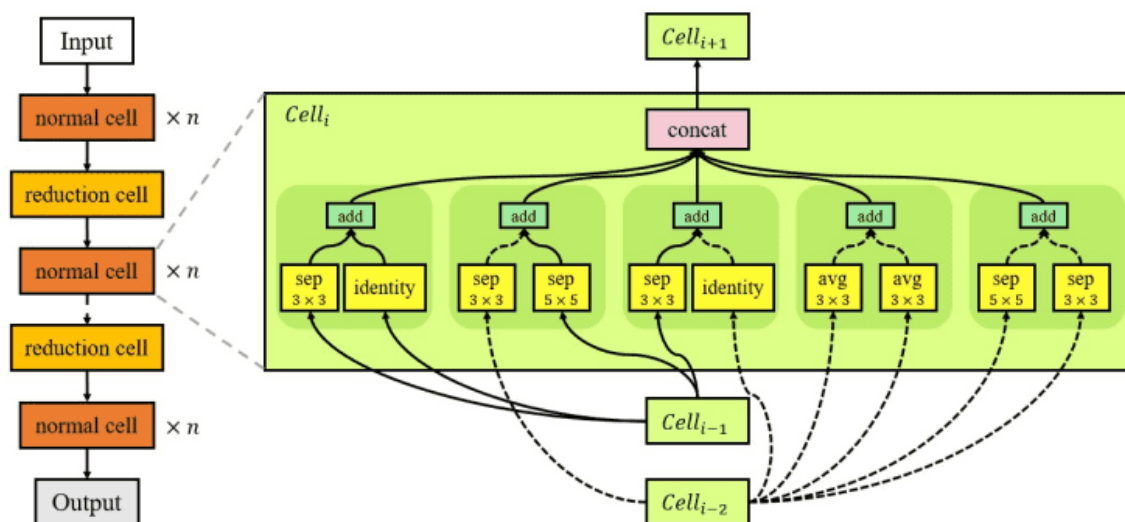
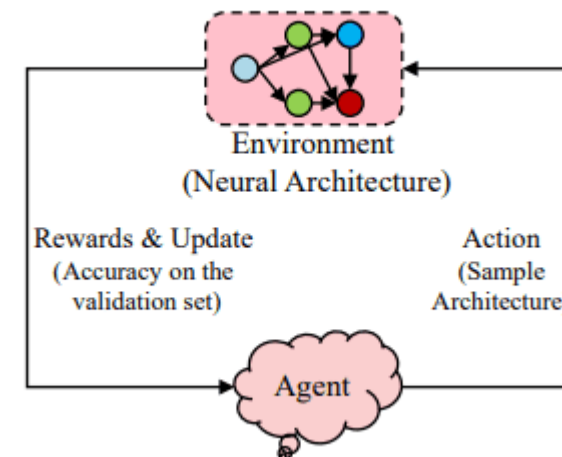
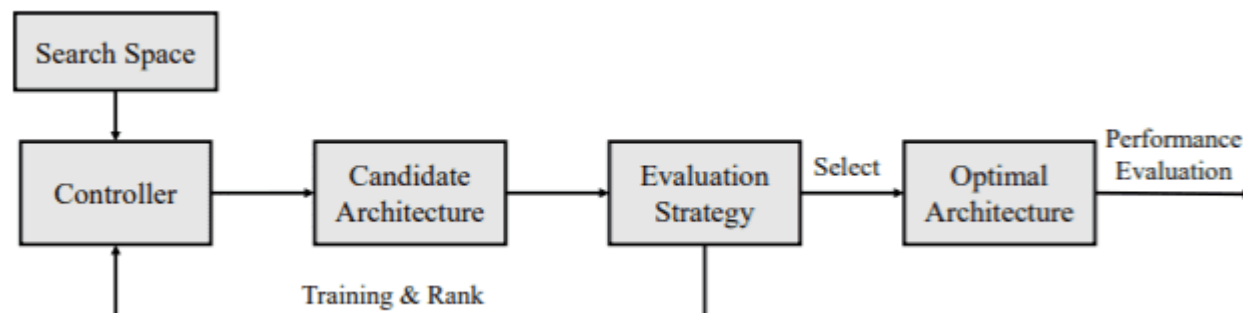
AutoML Frameworks





Поиск архитектуры модели

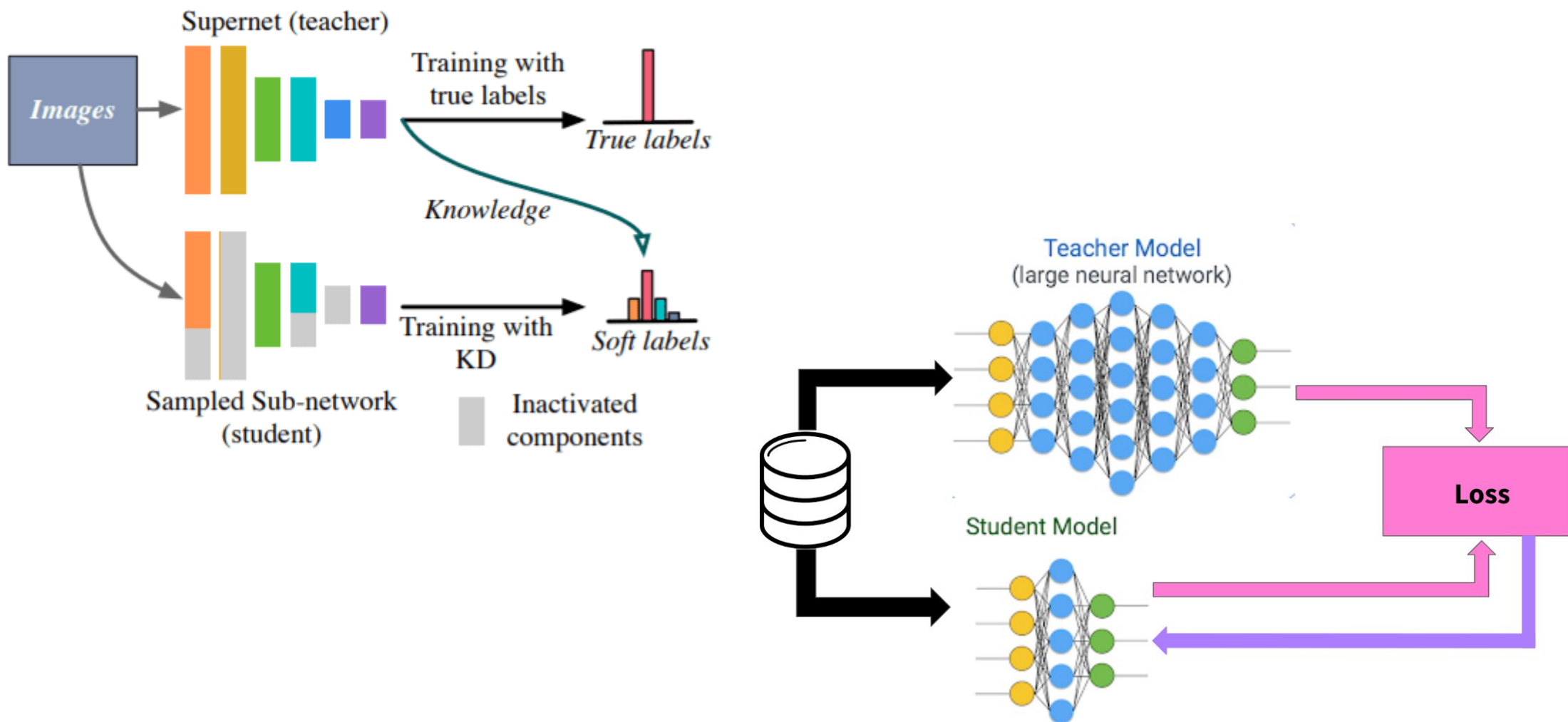
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Поиск архитектуры модели

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Группа по дисциплине:

<https://t.me/+8dShF1tFSDg0ZmJi>

